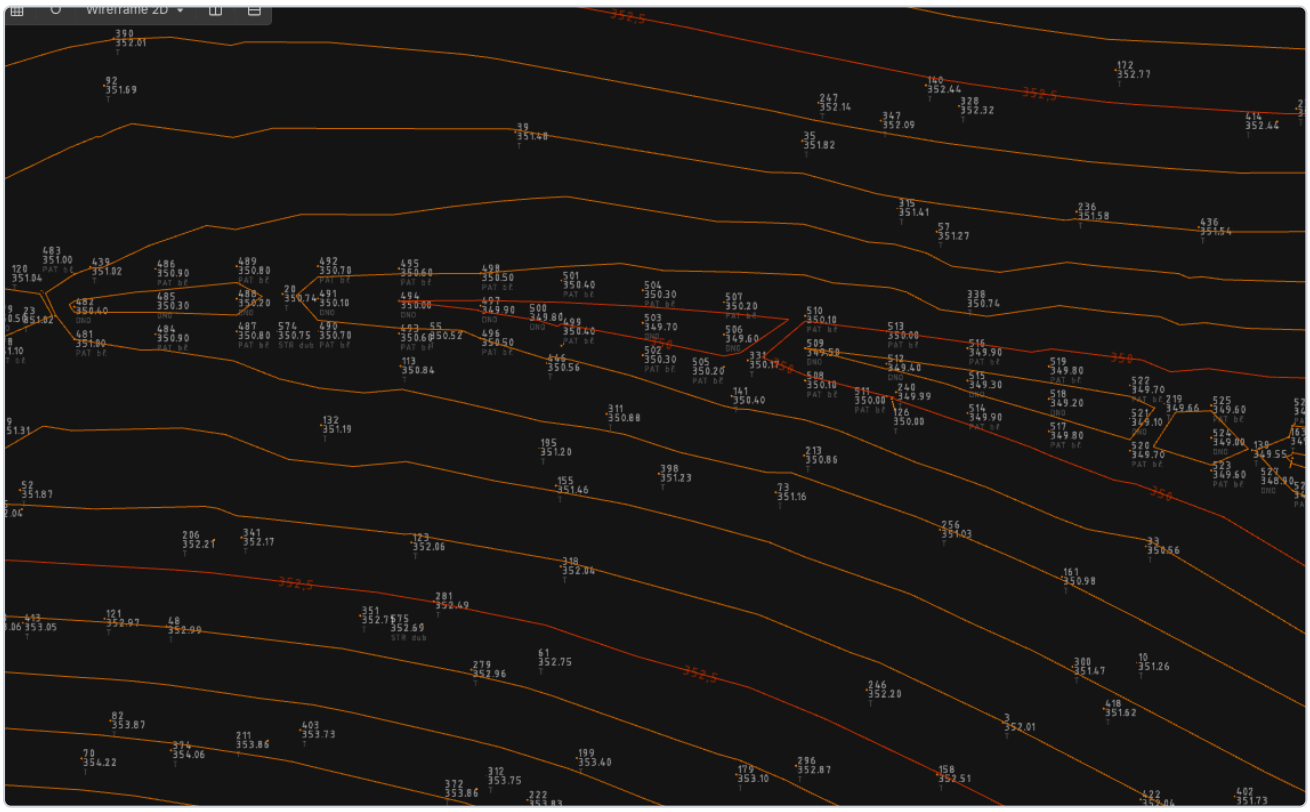
# Terén

Open CAD Studio plugin: survey points, TIN surfaces, breaklines, contours

Short manual · version 1.0.5



## 1. What Terén does

Terén builds a digital terrain model from survey points directly in an Open CAD Studio drawing:

- **Points** – import from text files (PNEZD, PENZD, PYXZD, NEZ, ENZ..., the format is detected automatically), markers and labels by styles, point groups, manual points, survey check.
- **Surfaces** – triangulated network (TIN) from point groups, breaklines, boundaries and holes, points from drawing objects, manual triangle edits, automatic recalculation.
- **Display** – contours with labels, triangles, outline, slope arrows; 3D viewer.
- **Export** – surface to LandXML or DXF (3D faces), group points to a text file.

Everything is stored in the drawing (DWG/DXF) – the drawing can be passed on and edited without the plugin; the plugin then reads the surface again.

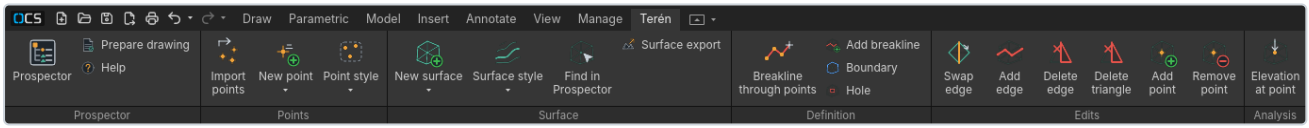
## 2. Installation

**Plugin Manager:** in Open CAD Studio open the Plugin Manager, find *Terén* and click Install. After a restart the **Terén** tab appears.

**Manually (Windows):** download `Teren-windows.zip` from the GitHub release, extract it and run `nainstalovat.bat`. The plugin is copied to `%APPDATA%\OpenCADStudio\plugins\cz.fluvio.teren`.

Each Terén release is built for one Open CAD Studio release – the Plugin Manager offers the right one; for a manual install pick the release from the table in the README (Terén 1.0.2 – 1.0.5 = Open CAD Studio 2026.45). The interface language (Czech / English) is set in Prospector › Settings.

## 3. Interface

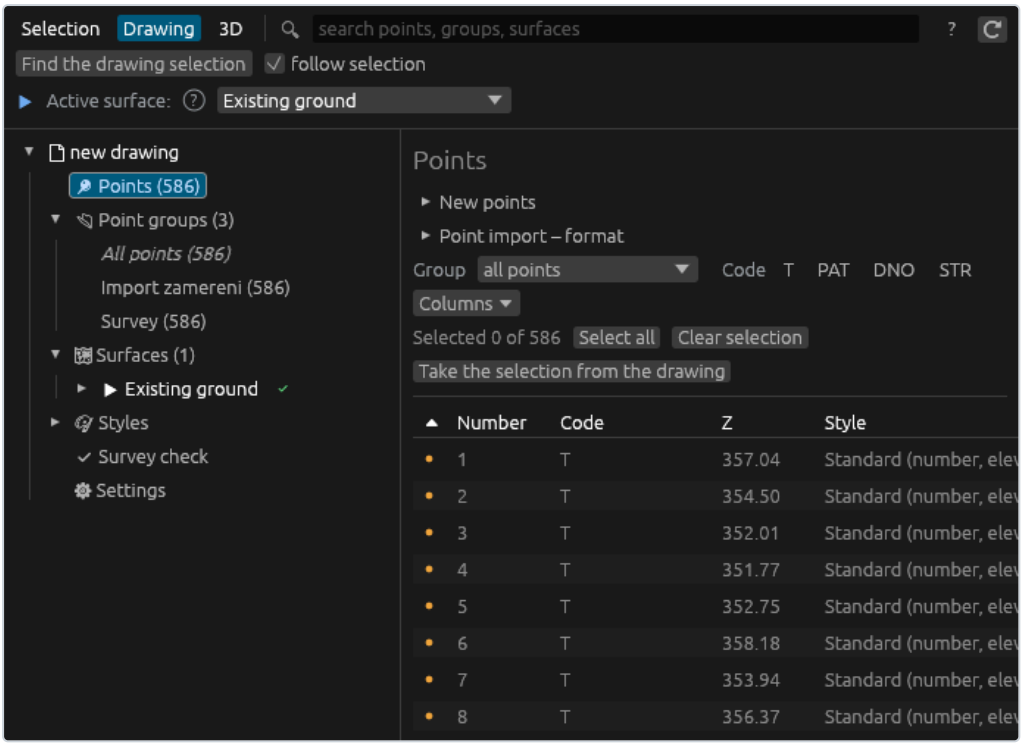



The Terén ribbon tab

| Group | Contents |
|---|---|
| Prospector | Prospector window, Import points, Drawing template (prepare the drawing from the template, save this drawing as the template), Help |
| Survey points | New point (manual, on surface), Point properties, Point style for selected points, Move points (on / off), Sync points, Reset labels, Group from selection |
| Surface | New surface (existing ground, design from 3D lines, from clicked points, wizard), Surface style, Recalculate surface, Show in Prospector, Surface: select by click / clicks pass to points |
| Surface definition | Breakline through points, Add breakline, Boundary, Hole |
| Triangle edits | Swap / add / delete edge, delete triangle, add point to surface / remove point from surface, undo surface edits |
| Outputs | Surface export, 3D viewer, Point export, Elevation at point, Survey check |

The **Prospector** is a separate window with three modes:

- **Selection** – what is selected in the drawing (point, surface) and what you can do with it: edit elevation and number, style, groups, recalculation.
- **Drawing** – a tree of everything the plugin has in the drawing: points (table), point groups, surfaces, styles, survey check, settings. Right-click opens a menu of actions.
- **3D** – surface viewer.

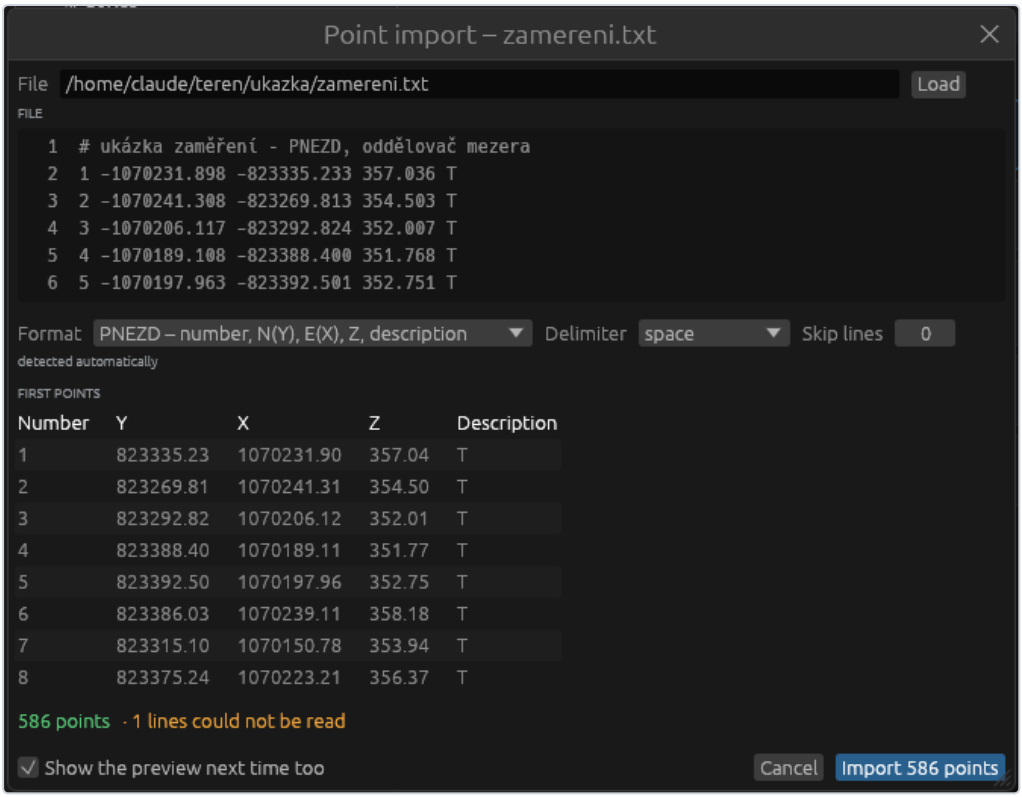


Prospector – Drawing mode, point table

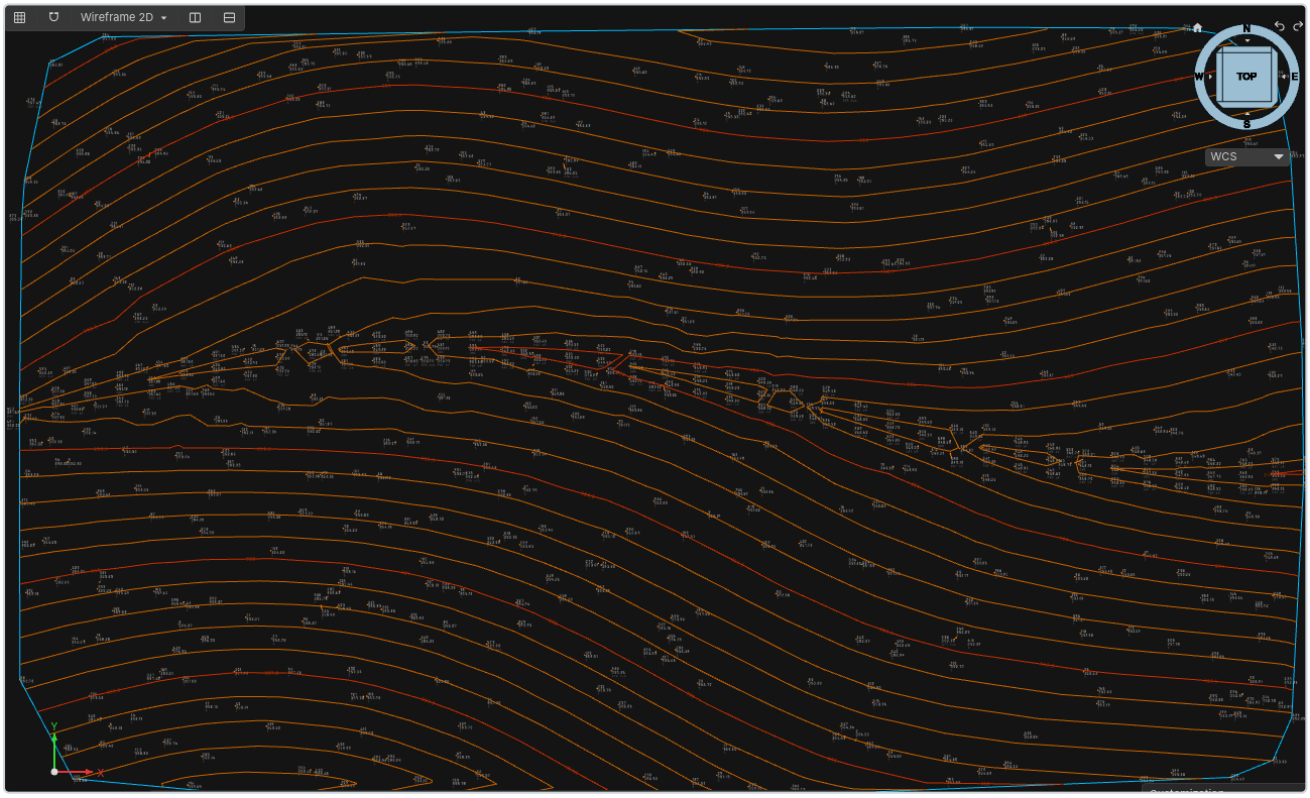
Help with workflows and a glossary is in the Prospector (**F1**); the ⓘ next to a term shows an explanation.

## 4. Ground from a survey – step by step

1. **Import points.** Terén › Import points, choose the file. The preview shows the first lines, the detected format and the first points; check the columns and click *Import*. S-JTSK points are drawn in negative drawing coordinates (positive Y, X in labels).
2. **New surface.** New surface › *Existing ground (from points)*. It takes the *Survey* group without STR codes (trees). The wizard (New surface › Wizard) lets you choose groups, max. edge length, style and contour interval.
3. **Breaklines.** *Breakline through points* – click survey points (slope edge, toe, top, ditch centreline); Enter = finish. Existing 3D polylines are added with *Add breakline*.
4. **Boundary.** Draw a closed polyline around the area, select it and click *Boundary*. *Hole* leaves out the inside of a polyline.
5. **Check and edit.** The *Triangles* style or the 3D tab shows the network. Swap or delete unsuitable edges with the Triangle edits tools; each edit is stored with the surface and reapplied on recalculation.
6. **Contours.** Right-click the surface in the Prospector › Style (e.g. *Contours with labels*) and Contour interval (0.1 / 0.5 m …).




Import preview




Surface with contours

A surface with **automatic recalculation** recalculates itself when points, breaklines or boundaries change. Otherwise the Prospector marks it ⚠ out of date and you just click Recalculate. Each Terén action (import, recalculation, redraw) is undone in one **Undo** step (Ctrl+Z).

## 5. Points

### Point styles

A style sets the marker (cross, circle, tree, manhole…) and the label (number, elevation, code). A manually set style applies, otherwise the style by code (the *for codes* field, e.g. STR → tree), otherwise the group's style, otherwise *Standard*. Sizes are in mm on paper and follow the drawing scale (annotation scale in the status bar, or Settings › Scale).

**Custom marker:** in the style editor choose *custom – drawing block* and select a block. The block's base point = point position, the block is scaled to the marker size; set the colour inside the block to ByBlock.

A point, its marker and labels stay together – clicking any part selects the whole point. **Moving a point:** grab the marker or a label and move it (with a grip or the Move command) – the whole point moves, its elevation stays and the surface is recalculated. With *moving a point label moves only the label* turned on in Settings, a moved label stays aside (with a leader line) and the point stays put; *Reset labels* in the point table brings it back. A copied marker or label is deleted. Overlapping labels are moved aside with a leader line (can be turned off in Settings).

### Moving points by grip, Terén point panel, cursor menu

**Move points** (Terén tab › Points; the button shows *on / off*, switch with its arrow) turns on custom grips: a circle appears on every point. Grab it and drag – the marker and labels move with the point; on release the point moves (its elevation stays) in one undo step. Click *Move points* again to turn the grips off (while they are on, a click on a point's centre starts a move instead of selecting it).

**Point properties** opens the *Terén point* panel in the dock: after selecting a point it shows the number, elevation, code and style. Edit the fields and click *Save*. With several points selected only the changed fields are saved (the number only for a single point).

The **cursor menu** (quad; turned on in Open CAD Studio with F12 or the QUAD field in the status bar) has a *Terén* tab for points and surfaces: point properties, move points, add point to surface, recalculate surface, Prospector.

### Manual points and editing elevation

*New point › Manual point*: click in the drawing or enter x,y; a window asks for the elevation (suggested from the surface or the previous point) and the code. *Point on surface* takes the elevation of the active surface. Change the elevation, number and code of an existing point directly in the point table: double-click a cell, type, Enter saves and moves to the next point (Esc cancels). For several selected points at once, set or add an elevation and set a code in the row above the table.

### Point groups

A group selects points by rules (code, numbers, import file), with exclusions and manually added / removed points. The preview shows in colour which points belong. A surface is built from groups – a change of a group recalculates the surface or marks it out of date.

### Survey check

Prospector › Survey check finds points whose elevation differs markedly from their surroundings (typo, point on a car). Suspicious points are circled on the TER_KONTROLA layer; one click leaves them out of the surface.

## 6. Surfaces

A surface is built from point groups, drawing objects (points, blocks, lines, 3D faces – e.g. a surveyor's drawing), breaklines, boundaries and holes, and manual edits. All are listed in the Prospector at the surface › *What the surface is built from*; each item can be disabled, shown or removed.

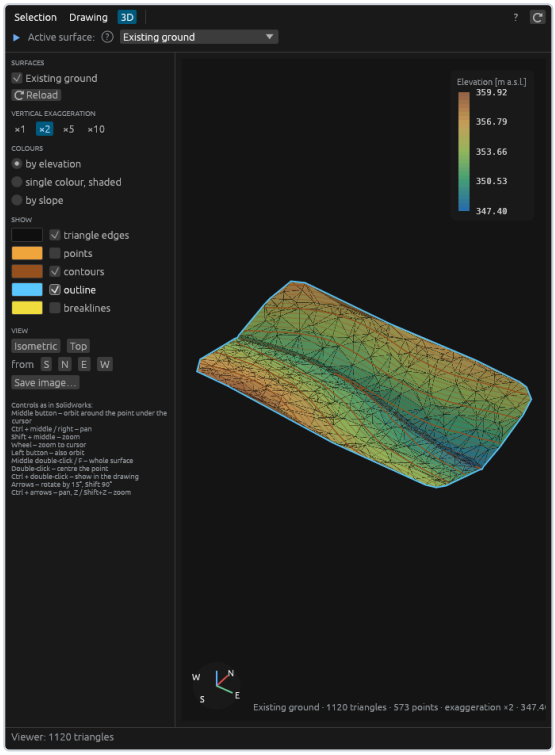
The **active surface** (top row of the Prospector, double-click in the tree) is used by the Triangle edits tools, Elevation at point and Point on surface.

The **surface style** sets what is drawn: triangles, contours (minor / major, colours, labels), outline, slope arrows. Edit styles in the Prospector under Styles; a built-in style can always be restored to its original settings.

In the drawing a surface is a block on its own layers; Settings › *Surface cannot be selected by clicking* lets clicks pass through to the points below.

## 7. 3D viewer

Colours by elevation, slope or a single colour, vertical exaggeration ×1 to ×10, contours, outline, breaklines, points and triangle edges – pick the colour of each element with the swatch next to it. Controls as in SolidWorks: the middle button orbits around the point under the cursor, Ctrl + middle pans, the wheel zooms, F shows the whole surface. Save image writes a PNG to the drawing folder.




## 8. Export

- **Surface export** (ribbon or right-click a surface): **LandXML 1.2** – exactly the same triangles, survey points and outline, readable by most design programs; **DXF** – each triangle as a 3D face.
- **Point export** of a group to a text file (PNEZD, PYXZD..., delimiter, UTF-8 or Windows-1250 encoding).

## 9. Settings and drawing template

Prospector › Settings: language (Czech / English, takes effect after Open CAD Studio is restarted), scale, label offset, import preview, excluded codes for a new surface, breakline snap, survey check threshold.

**Drawing template:** *Save this drawing as the template* stores the styles and scale in the user profile; *Prepare drawing* (ribbon) then sets up layers, point markers, scale and styles in a new drawing with one click.

## 10. Tips and troubleshooting

| Problem | Solution |
|---|---|
| No points appeared after import | Check the format and column order in the preview (S-JTSK: positive Y and X in the PYXZD format). |
| Long triangles across the edge of the surface | Set the Max. edge length or add a boundary. |
| Triangles cross a slope edge | Add a breakline (Breakline through points). |
| Trees are in the ground | Code STR is in the surface's *Exclude codes*; add other codes that do not belong to the ground. |
| The surface shows ⚠ out of date | Click Recalculate, or turn on automatic recalculation. |
| Labels are too large / small | Change the scale (status bar or Settings) – labels are redrawn. |